Gang Liu (Leo)



AI Algorithm Engineer / Forward-Deployed Engineer (FDE) — LLM Systems & Edge AI

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Education

Shanghai University of Engineering Science M.Eng. in Electronic Information (AI track) Sep 2024 – Jun 2027 (expected)

GPA: 3.36 / 4.0 Advisor: Prof. Xiang Liu Thesis: Multi-scale image detection for illegal-construction recognition Honors: First-Class Graduate Academic Scholarship

Guangling College, Yangzhou University B.Sc. in Information and Computing Science Sep 2017 – Jun 2021

GPA: 3.23 / 4.0 Honors: Outstanding Undergraduate Thesis Award

Technical Skills

• **LLM Inference Optimization & Edge Deployment**: LLM INT4 weight quantization (Qwen2-7B: 14 GB → 5.8 GB VRAM, 42 tokens/s single-GPU), QLoRA, vLLM, TensorRT FP16/INT8 operator fusion, ONNX Runtime, Jetson Orin Nano tuning, model pruning.

• **LLM / VLM & Agent Systems**: Qwen2 / Qwen2.5-VL multimodal fine-tuning, RAG (ChromaDB retrieval & rerank), Prompt Engineering, LangChain / LangGraph, multi-agent orchestration, MCP protocol, Function Calling.

• **Computer Vision & Deep Learning**: PyTorch, YOLOv8 / YOLOv11 (P2 pyramid, multi-scale fusion, decoupled head, instance segmentation, OBB), small-object detection, monocular ranging, ReID tracking, Kalman filtering, OpenCV.

• **AI-Native Engineering**: Claude Code, Codex, Antigravity, Grok, MiniMax, GLM, Kimi Code; FastAPI / React full-stack delivery; multi-agent agile workflow for end-to-end system builds.

Experience & Core Projects

Government-Enterprise AI Agent & RAG System — Zhuofan (current internship) | **LLM Agent / RAG** Sep 2026 – Present

Backend Development Engineer (AI Application Development Intern) [Stack: AI Agent / RAG / FastAPI / Function Calling]

• Contributed to core Agent logic: ReAct + workflow orchestration for intent recognition, task routing and result fusion; chained retrieval, pre-screening and business APIs via Tool Use / Function Calling.
• Built the policy RAG pipeline (ChromaDB + BM25 + Rerank); wrapped streaming output and session management in FastAPI (SSE) with timeout-retry, JSON Schema validation and human-handoff fallback.

Streetlight Detection & Monocular Ranging — CETC 50th Research Institute | **Production CV System** Oct 2025 – Jul 2026

Algorithm R&D Intern [Stack: YOLOv8 / Monocular Ranging / Kalman Filter / Industrial PC]

• Improved YOLOv8 with aspect-ratio-aware anchors/loss for slender pole targets (mAP50 92%+); monocular ranging via camera calibration + bottom-edge localization held relative error within 5% over 5–50 m.
• Added Kalman temporal filtering against vibration-induced false alarms; deployed and optimized on a vehicle-mounted industrial PC — live in production inspection fleets.

Multi-Agent Cybersecurity Threat-Analysis System | **National 3rd Prize · Team Captain** Jul 2025 – Oct 2025

Team Captain — full-stack architecture & LLM R&D [Stack: Qwen2-7B / INT4 Quantization / QLoRA / Multi-Agent / RAG]

• Led single-GPU lightweighting of Qwen2-7B: AutoGPTQ/AWQ INT4 quantization cut VRAM 14 GB → 5.8 GB (-58.5%) at 42 tokens/s offline; QLoRA fine-tuning on threat-intel corpora lifted term recognition +14.2%.
• Designed a 4-layer agent workflow (parsing → retrieval over 80k+ MITRE ATT&CK entries in ChromaDB → attack-chain reasoning → reporting); threat-assessment accuracy +12% vs. single-model baseline.

Master's Thesis — UAV Illegal-Construction Detection with Jetson Edge Deployment | **First Author / Lead** Oct 2024 – Present

Core of Master's research (Advisor: Prof. Xiang Liu) [Stack: YOLOv11s-seg / Jetson Orin Nano / TensorRT / Domain Adaptation]

• Built the RSIB-1K dataset (1,752 UAV images); re-engineered YOLOv11s-seg with a high-resolution shallow layer + DySample (+3.8% mAP50 small-target segmentation); domain adaptation with the WHU aerial dataset.
• Owned the PyTorch → ONNX → TensorRT FP16/INT8 pipeline on Jetson Orin Nano: 60% lower memory at 45+ FPS on-device, with DJI SDK streaming and GPS back-projection alerts.

understand_mov_v2 — Multimodal Video Retrieval & Understanding Platform | **Solo Full-Stack Project** Apr 2026 – Jun 2026

Independent design & development [Stack: Multimodal Retrieval / FastAPI / React / SQLite FTS5]

• Adaptive keyframe extraction + CLIP-style embeddings + multimodal LLM over a local vector store: natural-language video search with millisecond timestamp localization; FastAPI async pipelines, FTS5 + vector hybrid rerank, React front end.

Honors & Publications

• National Second Prize — China Mathematical Contest in Modeling (Huawei Cup), 2025 (Team Captain)

• National Third Prizes — Huawei Cup Post-Graduate Cybersecurity Competition 2025 & 7th Post-Graduate AI Innovation Competition 2025 (both as Team Captain); National IC EDA Elite Innovation Competition 2025 (Core Member)

• Provincial First Prize — CIMC Siemens Cup 2025; Provincial Second Prizes — CIMC 2026 & Post-Graduate EDC 2026

• Registered software copyright — RPA for Intelligent R&D-Expense Accounting V1.0 (sole developer); first-author manhole-cover detection paper (YOLOv11) under review at a Chinese core journal; thesis paper in preparation.